

Deep Learning

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1 Autoencoders

What is Unsupervised Learning?

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Definition 1

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Examples?

Autoencoders

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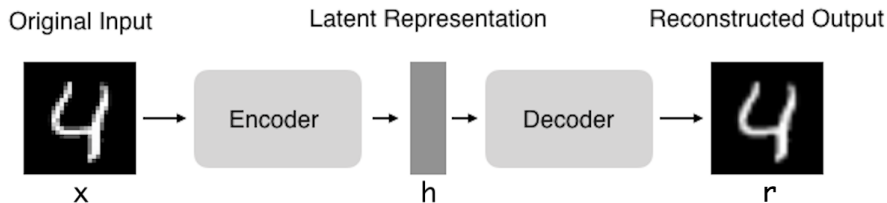
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This part aims to reconstruct the input from the latent space representation.

Autoencoders



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- Data denoising.
- In a lot of different tasks.

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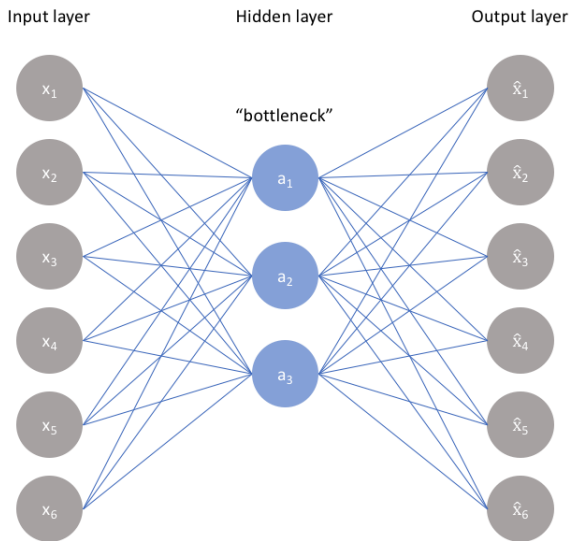
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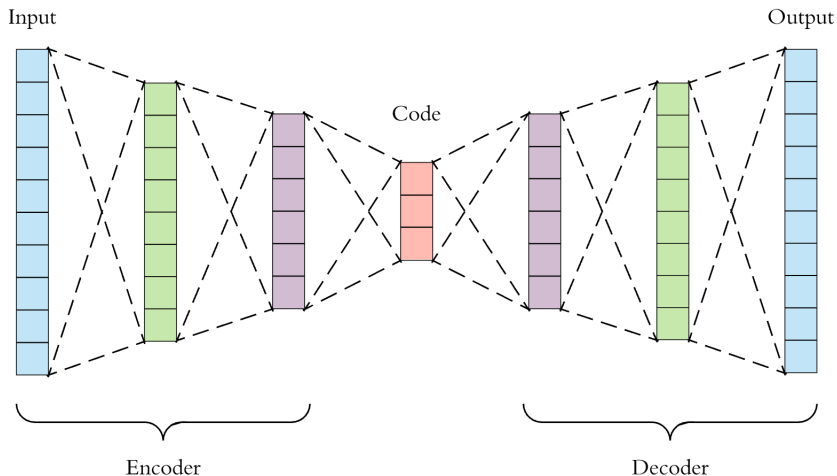
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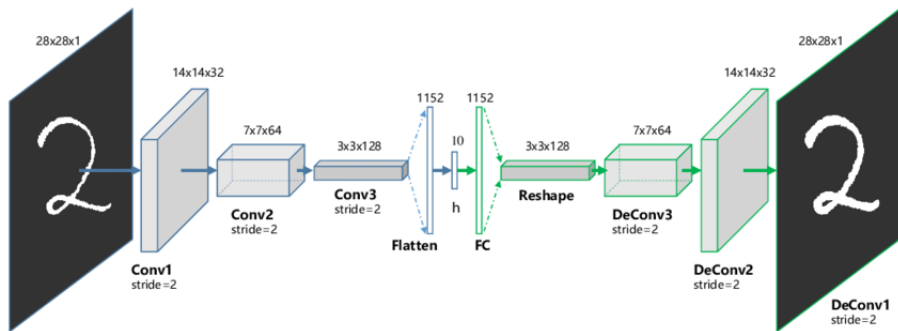
Vanilla Autoencoders



Multilayer/Deep Autoencoders



Convolutional Autoencoders



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- This is accomplished by adding a regularizer, or penalty term, to whatever cost or objective function the algorithm is trying to minimize.
- The end result is to reduce the learned representation's sensitivity towards the training input.

Contractive Autoencoders

Let f is our encoder, g is the decoder and D is our training dataset. In the previous cases we minimize this kind of loss function:

$$\sum_{x \in D} L(x, g(f(x))).$$

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In the case of contractive autoencoders we will minimize this one

$$\sum_{x \in D} \left(L(x, g(f(x))) + \lambda \|J_f(x)\|_F^2 \right),$$

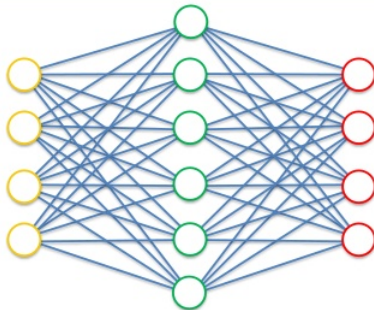
where the added summand is the square of Frobenius norm of the following Jacobian matrix:

$$[J_f(x)]_{i,j} = \frac{\partial f_j(x)}{\partial x_i}$$

i.e.

$$\|J_f(x)\|_F^2 = \sum_{i,j} \left(\frac{\partial f_j(x)}{\partial x_i} \right)^2.$$

Sparse Autoencoders



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- Sparsity penalty is introduced on the hidden layer. This is to prevent output layer copy input data. This prevents overfitting.

Sparse Autoencoders

Let f is our encoder, g is the decoder and D is our training dataset, which has n samples. Denote

$$\rho_j = \frac{1}{n} \sum_{x \in D} f_j(x).$$

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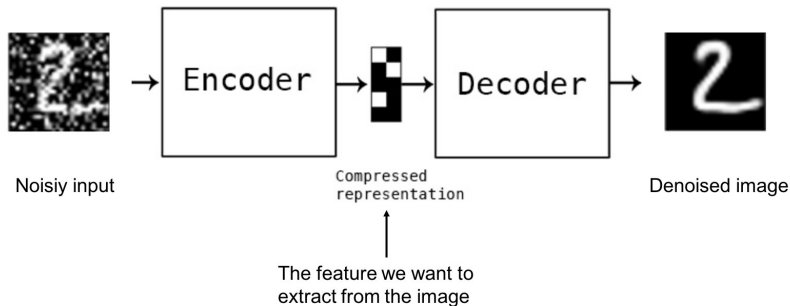
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where

$$KL(\rho || \rho_j) = -\rho \log \frac{\rho_j}{\rho} - (1 - \rho) \log \frac{1 - \rho_j}{1 - \rho}.$$

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- This helps to avoid the autoencoders to copy the input to the output without learning features about the data.
- The model learns a vector field for mapping the input data towards a lower dimensional manifold which describes the natural data to cancel out the added noise.

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Variational Autoencoders

- Can we generate content with autoencoders?
- Variational autoencoder can be defined as being an autoencoder whose training is regularised to avoid overfitting and ensure that the latent space has good properties that enable generative process.
- Instead of encoding an input as a single point, we encode it as a distribution over the latent space.

Variational Autoencoders

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- Let minimize the function

$$L(w) = KL(q_w(z|x) || p(z|x)).$$

Variational Autoencoders

Note that

$$KL(q_w(z|x) || p(z|x)) = \mathbb{E}_{q_w(z|x)} \left[\log \frac{q_w(z|x)}{p(z|x)} \right]$$

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$$\operatorname{argmin}_{w, w'} (KL(q_w(z|x) || p(z)) + \mathbb{E}_{q_w(z|x)} [\|x - f_{w'}(z)\|^2])$$

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Solution: we will assume that $q_w = \mathcal{N}(\mu, \Sigma)$ and $p = \mathcal{N}(0, I)$.

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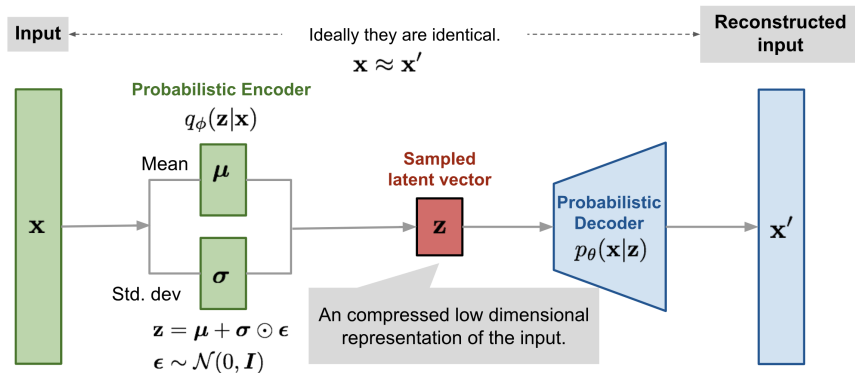
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Assumptions

- In practice, the encoded distributions are chosen to be normal so that the encoder can be trained to return the mean and the covariance matrix that describe these Gaussians.
- The distributions returned by the encoder are enforced to be close to a standard normal distribution.

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